

Erdős Problem #529

EP-529

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $d_k(n)$ be the expected distance from the origin after taking n random steps from the origin in $\mathbb{Z}^k$ (conditional on no self intersections) - that is, a self-avoiding walk. Is it true that $\lim_{n \rightarrow \infty} \frac{d_2(n)}{n^{1/2}} = \infty$? Is it true that $d_k(n) \ll n^{1/2}$ for $k \geq 3$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>